

Xiaomei Yuan
Guoyan Cheng
Thomas R. Malek

The importance of regulatory T-cell heterogeneity in maintaining self-tolerance

Authors' address

Xiaomei Yuan¹, Guoyan Cheng^{1,*}, Thomas R. Malek¹

¹Department of Microbiology and Immunology, Miller School of Medicine, University of Miami, Miami, FL, USA

Correspondence to:

Thomas R. Malek

Department of Microbiology and Immunology

Miller School of Medicine, University of Miami

1600 NW 10th Avenue R-138

Miami, FL 33136, USA

Fax: +1 305-243-5522

e-mail: tmalek@med.miami.edu

*Present address: Sylvester Comprehensive Cancer Center, Miller School of Medicine, University of Miami, Miami, FL, USA

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Summary: CD4⁺ Forkhead box protein 3 (Foxp3)⁺ regulatory T cells (Tregs) are the major cell type that mediates dominant tolerance in the periphery. Over the past decade, extensive study of Tregs has revealed that these cells express substantial heterogeneity to maintain tolerance and regulate immune responses. Tregs possess heterogeneity with respect to their origin and processes for development, functional activity, migratory pattern, and activation status. Some of the same environmental cues and molecular pathways utilized to generate specialized T-effector cells are also integrated by Tregs to colocalize and fine-tune suppressive mechanisms to optimally regulate and restrain distinctive self and antigen-specific T-cell responses. Here, we review our current understanding and significance of Treg heterogeneity in maintaining peripheral immune tolerance. We also highlight recent work from our laboratory that has studied the extent phenotypically distinct Treg subsets are related to each other and expand in an ordered fashion to give rise to highly activated short-lived Klrp1⁺ suppressor cells to optimize immune regulation and maintain homeostasis of the Treg compartment.

Keywords: Tregs, tolerance, subsets, effector, memory, Klrp1

Introduction

CD4⁺ Forkhead box protein 3 (Foxp3)⁺ regulatory T cells (Tregs) were initially described as thymus-derived T cells that suppress and normally keep in check peripheral autoreactive T cells that escape thymic negative selection (1). Foxp3 is the signature transcription factor of Tregs that promotes the Treg gene suppressive program while repressing key genes characteristic of T-effector (Teff) cells such as interleukin-2 (IL-2) (2–8). When Tregs are absent as a consequence of effective deletion or impaired Foxp3 expression, severe rapid lethal multi-organ autoimmune attack occurs (4, 5, 7–10). Thus, the sum of all other immune regulatory mechanisms is ineffective in controlling autoreactive T cells in the absence of functional Tregs. Intensive study has revealed that Tregs are also important in regulating many other immune responses, including responses to pathogenic agents, environmental allergens and food, gut microbiota,

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tumors, and transplanted tissues, which has been reviewed elsewhere (11–17). The high T-cell receptor (TCR) diversity of thymic-derived Tregs (tTregs) provides a range of specificities to respond to self- and non-self antigens (18–20). In addition, under the proper environmental conditions, conventional peripheral CD4⁺ T cells can be induced to express Foxp3 and function as suppressor cells (pTregs). These Tregs provide another level of regulation by which various antigen-specific immune responses are regulated and constrained.

Expression of Foxp3 represents a unifying feature of tTreg and pTregs. Nevertheless, it is apparent that Tregs exhibit substantial heterogeneity besides the origin of their development to control immune responses to self and foreign antigens. This review discusses the cellular and molecular basis that contribute to developmental, functional, migratory, and activation-related heterogeneity of Tregs and their importance for proper suppressive function. We also summarize recent studies that begin to assess the inter-relationship between activated and memory Tregs and our work on phenotypic distinct Treg populations, including the role of Klrp1⁺ Tregs in immune suppression and homeostasis. These studies are generally consistent with the notion that many properties of Tregs are expressed by co-opting activities and mechanisms that regulate Teff immune responses. Tregs can be viewed as another subset of activated T cells whose role is to regulate and suppress, rather than to induce and promote, immune responses.

Developmental heterogeneity of Tregs

The large majority of Foxp3⁺ Tregs that are found in the periphery arise as a consequence of development into the Treg lineage in the thymus (tTregs). After positive selection and development of CD4 single positive thymocytes, tTreg commitment occurs primarily in the medulla through selection events on medullary epithelial and dendritic cells (21, 22). These selection events result in tTregs that express a diverse TCR repertoire with a relatively high affinity for self-antigens that only partially overlaps with that found on conventional CD4⁺ T cells (18–20, 23–26). The developmental cues that promote tTregs have been recently reviewed (27). In brief, relatively high TCR signaling in conjunction with costimulatory signaling results in activation of nuclear factor of activated T cells (NF-AT), activator protein-1 (AP-1), and CARMA1/Bcl10/Malt1-dependent activation of NF-κB, which directly acts on the Foxp3 gene to induce Foxp3 expression (28–32). Full maturation of

tTregs requires further activation from transforming growth factor-β (TGFβ)-dependent Smad3 and interleukin-2 (IL-2)-dependent signal transducer and activator of transcription 5 (STAT5) that provides survival and growth signals and further acts on the Foxp3 gene to induce heightened levels of Foxp3 required for the expression of the Treg functional program (33–42). Transcriptional regulation of Foxp3 critically depends upon factors that exert their activity on its promoter and conserved non-coding sequence 2 (CNS2) and 3 (CNS3) (32). Mature tTregs seed peripheral immune tissues and are critically required to suppress peripheral autoreactive T cells that were not deleted during thymic negative selection.

Under appropriate environmental cues, conventional CD4⁺ T cells can develop in the periphery into Foxp3⁺ Tregs (pTregs). These pTregs regulate antigen-specific immune responses and contribute to tolerance in mucosal tissues and during pregnancy (43, 44). Along with a requirement for TCR and costimulatory signaling, the development of pTregs is highly dependent upon TGFβ (45–48). The mucosa is a preferential site for pTreg induction, in part due to abundant TGFβ and retinoid acid produced locally by epithelial cells, macrophages, and CD103⁺ DCs (49–54). The development of pTregs is particularly dependent on TGFβ-activation of pSmad3 through its action on CNS1 of Foxp3 (32). Accordingly, after germ-line deletion of Foxp3 CNS1, mice lack pTregs and exhibit Th2-type inflammation in the gut and lungs (43). This finding supports the notion that pTregs provide a critical role for mucosal tolerance.

Another factor contributing to the generation of pTregs is commensal bacteria and food antigens in the gut and allergens in lung airways (46, 54–57). These environmental antigens select TCRs with specificities that are unique to the mucosa (58). Some of these specificities are likely derived from conventional T cells because when individual TCR specificities from the gut mucosa were expressed in bone marrow chimeras, they were not selected in the pool of tTregs. In another study, some tTregs, however, have been shown to respond to microflora-derived antigens (59). Thus, mucosal Tregs are comprised of a mixture of tTregs and pTregs, and the activity of both is likely required for effective tolerance. In this regard, the most effective means to prevent autoimmunity associated with Foxp3-deficient mice required the transfer of large number of tTregs and some conventional CD4⁺ T cells, which developed into pTregs (60).

Although it is commonly accepted that the **peripheral conversion** of activated T cells to **Tregs** contributes to **peripheral tolerance**, the **proportion** of **tTreg** and **pTreg** within mucosal sites has been **difficult to** precisely **determine** by **TCR** repertoire analysis. Another approach is to identify a marker that definitively **distinguishes** **tTregs** and **pTregs**. Two candidates are Helios and neuropilin-1, as these molecules are highly expressed by **tTregs** but poorly expressed by **in vitro**-induced Tregs (**iTregs**) and by **pTregs** in a number of settings (61–63). These two markers, however, do not perfectly **distinguish** **tTregs** from **iTregs** and **pTregs**. For example, many **pTregs** in the central nervous system expressed neuropilin-1 due to inflammation during spontaneous chronic experimental autoimmune encephalomyelitis (62). Likewise, expression of Helios is sometimes found on **iTregs** and **pTregs**, depending on the type of antigen-presenting cells and activation signals encountered (64–66).

Recent work supports the notion that **tTregs** are a **stable lineage** of **dedicated suppressor** cells with **minimal** capacity to **de-differentiate** and **convert into** **Teff** cells. The **Foxp3** promoter and the **CNS2** are highly demethylated in **tTregs**, which contribute to their **lineage stability** (32, 67–70). In contrast, **iTregs** have **Foxp3** **CNS2** that is substantially **methy-lated** and convert to **Th17** cells under conditions of lower **TGF β** and increased inflammatory signals (67, 68, 71). **In vivo** **pTregs** have also been shown to lose expression of **Foxp3** and **convert to** **Th** cells (70, 72–74). Considering that **tTregs** behave as a **stable lineage**, this plasticity likely reflects the **instability** of **pTregs**. Such plasticity might provide a mechanism to help resist pathogenic infections.

Functional **heterogeneity** of **Tregs**

Early considerations of Treg **suppressive** function focused on the hypothesis that there was a single major dominant mediator of **suppression**. However, extensive study of Tregs has revealed that Tregs express **a number of distinct mediators that downregulate immune responses**. Some of these mediators include IL-10, **TGF β** , IL-35, granzyme, Galectin-1, cytotoxic T-lymphocyte antigen-4 (CTLA4), CD39, CD73, neuropilin-1 (Nrp-1), lymphocyte activation gene-3 (LAG-3), and fibrinogen-like protein-2 (FGL-2). Detailed discussion of their role in Treg-mediated **immune suppression** has been reviewed recently (75–77). **In vivo**, these mediators create a localized immune **inhibitory** environment in part by directly acting on autoreactive T cells and in part by promoting **tolerogenic** DCs. **Suppressive** activity of Tregs

extends beyond **adaptive immunity**, because Tregs also **suppress innate immune** cells, such as mast cells, macrophages, and NK cells (75, 78).

The expression of multiple mediators raises the question concerning the extent whether an individual mediator is dominant versus multiple **mediators** co-operating to **suppress** immune **responses**. The knockout of several of these mediators only in **Tregs** provides some answers to this question. Mice engineered to contain **Tregs** that do not express **CTLA4** exhibited severe and lethal autoimmunity, even though **Treg** development and **peripheral** numbers were normal (79). Thus, expression of **CTLA4** by Tregs is essential for normal immune **homeostasis** and **tolerance** and represents a core mechanism of the Treg **suppressive** program. In contrast, mice that contain Tregs engineered not to express IL-10 spontaneously developed inflammatory bowel disease at 3–6 months of age and exhibited lung and skin hypersensitivity (80). These findings point to an important role for Treg production of **IL-10** to suppress certain tissue specific responses, but the generally long time frame to obvious pathological changes suggests that other Treg **suppressive** mechanisms are also active but **not sufficient** to fully control these unwanted immune responses.

Current studies favor a view that **Tregs** possess **functional heterogeneity** and **specialization** to **suppress distinct types** of immune responses and to adapt and integrate signals from the local environment present in **distinct** tissues. One example already discussed is the gut mucosa that is dominated by IL-10-producing highly activated **Tregs** and that is permissive for development of **pTregs** to maintain **tolerance** to commensal microbiota and food antigens. A more recent example is the detection of a specialized population of Tregs in adipose tissue. Genome-wide profiling of Tregs from adipose tissue confirmed their expression of the Treg gene signature but also **detected distinct** sets of genes associated with a **unique** trafficking, anti-inflammatory and lipid metabolism profile (81). Depletion of Tregs in fat tissue correlated with increased inflammation and insulin resistance. Reciprocally, expanding Tregs in this tissue increased the **anti-inflammatory** cytokine **IL-10** and decreased glucose levels in the blood. The unique property of **Tregs** in adipose tissue is controlled in part by PPAR- γ , a key regulator of adipocyte differentiation (82). PPAR- γ not only determines the **distinctive** phenotype of fat Tregs but also regulates local Treg homeostasis. These findings link the failure of specialized **Treg** activity to a new and alternative mechanism to explain how chronic inflammation contributes to obesity.

There are also substantial additional data that Tregs respond to the local milieu and behave as subtypes that are equipped with distinctive suppressive programs to combat immune responses, in particular, Th1, Th2, and Th17 responses by CD4⁺ T cells (Table 1). Various cytokines and transcriptional regulators are linked to specific Th activities, e.g. IFN γ activates STAT1 and IL-12 activates STAT4 to drive T-bet expression for Th1 cells, IL-4 activates STAT6 to drive Gata3 and IRF4 expression in Th2 cells, and IL-6 and TGF β activates STAT3 to drive ROR γ t expression for Th17 cells (83). Tregs also utilize these same transcriptional regulators to generate suppressive responses to counteract the respective Th responses, which we have designated here as Ts1, Ts2, and Ts17 in Table 1. Correspondingly, Tregs with deletion in T-bet, IRF4, and STAT3 showed selective inability to control Th1, Th2, and Th17 immune responses, respectively (84–86).

The cytokines promoting lineage-specific suppressor activity of Tregs are largely distinct from the pro-inflammatory cytokines that promote Th subsets (87–89) (Table 1). These cytokines, nevertheless, also generate signals that converge on the same transcriptional regulators used by Th cells to activate T-bet, IRF4, and STAT3, leading to expression of CXCR3, CCR8, and CCR6 on Tregs, respectively. This type of control colocalizes the Tregs to appropriate tissue sites to regulate Th1, Th2, and Th17 responses. Along with regulation of homing, these transcriptional regulators also distinctively control the expression of molecules associated with suppressive activity of Tregs. During a Th1 response, optimal expression of CTLA4, IL-10, and TGF β was dependent

upon T-bet (85). Targeted deletion of IRF4 in Tregs resulted in developing severe Th2 autoimmunity (86). Although Treg expression of CTLA4, TGF β , and GITR was unchanged, other Treg functional molecules, including Fgl2, IL-10 and granzyme B, were decreased, linking Treg expression of IRF4 to specific aspects of the Treg gene program and to regulate Th2 responses. Mice with targeted inactivation of STAT3 in Tregs were associated with impaired function to suppress Th17 cells and developed severe intestinal inflammation (84). Gene expression profiling revealed that STAT3-deficient Tregs expressed lower levels of IL-10, granzyme B, Ebi3, and perforin, linking these suppressive mediators to control Th17 responses.

T-follicular helper (Tfh) cells are CD4⁺ T cells specialized in regulating the germinal center reaction and their function depends on relatively high expression of Bcl6 and CXCR5; the latter is necessary for Tfh cell trafficking into the B-cell follicles (90–93). In Tregs, the transcription factor Bcl6 is also critical for the expression of CXCR5 and is essential for controlling the germinal center reaction (94–96). Tregs lacking either CXCR5 or Bcl6 showed impaired suppressor function and caused enhanced germinal center responses characterized by pronounced antibody production, affinity maturation, and accumulation of plasma cells. Collectively, all these data indicate substantial Treg heterogeneity develops as Tregs respond to antigens in localized environments that also vary with regard to cellular and inflammatory signals.

Phenotypic heterogeneity contributes to Treg migration

Besides the above-mentioned relationship to varied chemokine receptor expression by Th and Treg populations, there is also additional heterogeneity in expression of other chemokine receptors that regulate Treg trafficking. A two-step switch model of chemokine receptor expression has been proposed to depict the migration of Tregs from the thymus to peripheral lymphoid and non-lymphoid tissues (97). The first switch occurs in the thymus where Tregs acquire CCR7 expression for homing to secondary lymphoid tissues. Tregs emerging from the thymus are uniformly CD62L^{hi} CCR7⁺ CXCR4^{low}, which promotes migration to secondary lymphoid organs. The second switch occurs in secondary lymphoid tissues where Tregs are educated by local antigenic stimulation to express an array of chemokine receptors and integrins, e.g. CCR2, CCR5, CCR6, CXCR6, CCR4, CCR8, and α 4 β 7, while expression of CD62L and

Table 1. Comparison of Th and Treg subtypes

	Cytokine	Transcription factor	Function
Th1	IFN- γ , IL-12	T-bet	Production of IFN- γ ; expression of CXCR3 on T cells
Ts1	IFN- γ , IL-27	T-bet	Differentiation of CXCR3 ⁺ Tregs; high expression of GITR, CTLA4, CD103, IL-10, TGF- β on Tregs
Th2	IL-4, IL-2	Gata3, IRF4	Production of IL-4, IL-5, IL-13, IL-10 cytokines; Ig production
Ts2	?	IRF4	Upregulate Icos, Fgl2, IL-10, Gzmb, maf, Ccr8, IL-1RI on Tregs
Th17	IL-6, TGF- β	ROR γ t	Production of IL-17A, IL-17F, IL-21, IL-22, GM-CSF
Ts17	IL-10	STAT3	Upregulate IL-10, Ebi3, Gzmb, Prf1, Ccr6 expression on Tregs
Tfh	IL-6, IL-21	Bcl6	Expression of CXCR5 on T cells; germinal center formation; affinity maturation
Tfs	?	Bcl6	Expression of CXCR5

CCR7 is downregulated, which promotes trafficking to various non-lymphoid tissues. Generally, CD62L^{lo} Tregs found in secondary lymphoid tissues express increased levels and diversity of the trafficking molecules, which provide an advantage over conventional T cells to migrate to non-lymphoid tissues to maintain local immune homeostasis.

The types of chemokine receptors expressed by Tregs are influenced by the local environmental cues associated with the regional lymph nodes. For example, expression of CCR4 in peripheral LNs is associated with trafficking to skin and lung, while expression $\alpha 4\beta 7$ and CCR9 in the MLNs is associated with homing to the small intestine (98–100). However, homing to the gut mucosa is complicated, because this process involves lymphocyte entry into the lamina propria and across the basement membrane to the intra-epithelial lymphocyte compartment. For example, in mice deficient in CCR9 or its ligand CCL25, homing of CD8 $\alpha\alpha$ TCR $\gamma\delta$ cells to the intraepithelial lymphocyte compartment of small intestine was impaired, but homing of T cells, particularly CD4⁺ T cells, to the lamina propria was unaffected (101). Other chemokine receptors, such as CXCR3 and CXCR4, likely complement CCR9/CCL25 to facilitate lymphocyte homing to the gut mucosa (102–104). In addition, integrin $\beta 7$ and αE (CD103), which form the $\alpha E\beta 7$ heterodimer, may also contribute to T-cell homing or retention in the gut (105–108). However, the absence of integrin $\beta 7$ largely reduced Treg trafficking to the gut mucosa but Tregs still suppressed colitis, suggesting that suppression at least in part may occur in draining lymph nodes (109). One molecule that may be essential for Treg homing to the large intestine is GPR15 (110). In this regard, Tregs that do not express GPR15 do not efficiently home and prevent colitis. Unlike small intestine tropic homing molecules, which are regulated largely by retinoic acid, GPR15 is TGF β -dependent and regulated by colonic microbiota, suggesting that the local environment endows Tregs with unique properties that at least in part affect their homing.

Heterogeneity reflecting effector and memory Treg subsets

The discovery early on that CD25 (IL-2R α) was an excellent, albeit not perfect, marker for Tregs indicated that Tregs express properties of activated T cells (111). When considering Foxp3 expression, which more precisely marks Tregs, typically 70–80% of Tregs are CD25⁺, firmly establishing the activated phenotype of Tregs. In vivo Tregs express higher levels of CD25 than conventional activated T cells, and this

in part reflects that fact that Foxp3 functions to promote CD25 transcription. Most Tregs not only express CD25 but also express CD122 (IL-2R β) and CD132 (γc) required for the high affinity IL-2R, which is characteristic of activated T cells. Tregs stringently depend on IL-2 during thymic development and subsequent maintenance in the periphery (112). IL-2R signaling in Tregs results in a classical positive feedback loop with Foxp3, where IL-2-induced STAT5 directly contributes to promote Foxp3 transcription and Foxp3 in turn directly promotes CD25 transcription (4, 37, 39, 40). This positive feedback helps to enforce the Treg suppressive program, which critically depends on Foxp3. In this regard, the functional activity of Foxp3⁺ CD25^{neg} Tregs is unstable and such cells are prone to lose Foxp3 expression (70, 72, 73).

Constant stimulation with self-antigens, costimulatory molecules, and IL-2 represent important drivers for Tregs to express an activated phenotype. Indeed, Tregs are also characterized by expression of many other activation markers besides the IL-2R, including CD69, CD38, CTLA4, 4-1BB, OX40, CD62L^{lo}, CD45RB^{lo}, GITR, and CD103 (113, 114). Similar to CD25, some of the molecules, e.g. GITR, CTLA4, 4-1BB, CD103, and OX40, are expressed at higher levels on Tregs than conventional T cells, and this property likely endows Tregs with a competitive advantage over autoreactive T cells for accessing their respective ligands, providing a mechanism that would favor immune suppression and tolerance over autoimmunity.

Some markers of T-cell activation on Tregs readily discriminate subpopulation of cells. One that has received considerable attention is CD103, which is expressed on 10–30% of Tregs in secondary lymphoid tissues but is found on 75–85% of Tregs within some tissue sites such as the skin and gut mucosa (115). Gene expression profiling is consistent with CD103⁺ Tregs favoring migration into tissue sites and expressing enhanced suppressive activity due to high levels of Treg suppressor molecules (116). Consistent with this finding, CD103⁺ Tregs showed increased suppressor function when compared to CD103[−] Tregs in vitro and showed robust suppressive activity when tested in the T-cell-transfer model of colitis (114, 117, 118). In addition, CD103⁺ but not CD103[−] Tregs suppressed acute arthritis, consistent with their important role in immune regulation within tissue sites (119). Upon transfer into a lymphopenic environment, many Tregs adopted a CD103⁺ phenotype and expressed a transcriptional profile that resembled gut Tregs, where most Tregs are CD103⁺ (116). Thus, the development of CD103⁺

Tregs in part depends upon high proliferation reflected as a consequence of lymphopenia or frequent gut antigenic stimulation. All these findings support the notion that CD103 marks a specialized subset of effector Tregs that undergo substantial proliferation. However, besides its role as an adhesion molecule to E-cadherin, it is not known whether CD103 is actively required for suppressive function of this Treg subset.

At the transcriptional level, IRF4 and Blimp-1 are key transcription factors involved in promoting a subpopulation of highly activated effector-phenotypic Tregs (120). Blimp-1⁺ Tregs are characterized by high expression of CD103, CD44, inducible costimulatory (ICOS), glucocorticoid-induced tumor necrosis factor receptor (GITR), and low expression of CD62L. The expression of Blimp-1 is increased in Tregs stimulated with IL-2 and further upregulated in the presence of inflammatory cytokines, including IL-12, IL-6, and IL-4, consistent with Blimp-1 in regulation of a specialized Treg subpopulation. In the absence of Blimp-1, highly activated Tregs are still readily detected, indicating that Blimp-1 is not required for the development of these effector-phenotypic Tregs, but some of their other activities are impaired, such as production of IL-10. In addition, Blimp-1 regulates Treg homeostasis in mucosal sites. In contrast, these highly activated Tregs are substantially reduced in IRF4-deficient mice, indicating that the development of effector-phenotypic Tregs depends on this transcriptional regulator, which is also important for Th2 immune responses. IRF4 and Blimp-1 have been shown to control a subset of overlapping genes, but IRF4 is required for normal Blimp-1 expression by Tregs. Collectively, these data indicate that the activity of IRF4 is upstream of Blimp-1 in the development of highly activated Tregs.

The pattern of expression of CD44, CD62L, and CD127 (IL-7R α) by conventional T cells is useful to distinguish naive (CD44^{lo}, CD62L^{hi}, CD127^{hi}), effector (CD44^{hi}, CD62L^{lo}, CD127^{lo}), effector/memory (Tem) (CD44^{hi}, CD62L^{lo}, CD127^{hi}), and central memory (Tcm) (CD44^{hi}, CD62L^{hi}, CD127^{lo}) T cells. For Tregs, most are CD44^{hi} and CD127^{lo}, consistent with activated effector cells, although there is heterogeneity in CD62L expression, which is important with respect to the lymphoid versus non-lymphoid localization of Tregs, as discussed above. Notably, there are very few Tregs that are obviously Tem and Tcm based on these markers. However, recent studies are consistent with the presence of a persistent Treg memory pool that is driven by antigen. A sophisticated triple transgenic model was developed where transgenic ovalbumin (OVA)-specific

conventional CD4⁺ T and Treg cells could be followed over time in response to transient expression of OVA in the skin (121). Upon an initial expression of OVA, a pathogenic response toward the skin by the conventional TCR transgenic effector cells was noted that was eventually suppressed by the TCR transgenic Tregs, which showed an activated phenotype. Importantly, in this model system, the transient nature of OVA expression eventually resulted in an OVA-free environment, yet the OVA-specific Tregs persisted in the skin. Upon re-induction of OVA, only a mild form of skin disease was noted and was accompanied by activation, proliferation, and suppression by the OVA-specific persistent Tregs. This study indicates that Tregs exhibit properties of memory cells, i.e. persistence in the absence of antigen and a faster response upon recall. More recently, it has been shown in this system that IL-2 is required to generate effector Tregs, while IL-7 is required for their persistence (122). Evidence for memory Tregs has also been associated with pregnancy (123). Maternal Tregs contribute to maternal–fetal tolerance. Importantly, after delivery, fetus-specific Tregs persisted, and during a second pregnancy they rapidly proliferated in response to the same fetal antigens and promoted tolerance.

The inter-relationship of Treg subsets

Although phenotypic and functional heterogeneity of Tregs are well characterized, the potential relationship between these distinct Treg populations has not been extensively studied. Our laboratory has begun to directly address this issue. One underlying tenet for these studies is that distinct expression of key surface proteins by Tregs marks varied activation states derived from a common precursor as Tregs respond to auto-antigens. This idea implies that Tregs adopt a similar progression upon encountering self-antigen, costimulatory, and cytokine signals to move from a relatively quiescent state to a proliferative and more activated state to yield increased number and potency of suppressive Treg cells. Eventually many of the activated Tregs undergo apoptosis to maintain Treg homeostasis, but some Tregs persist as memory cells (Fig. 1). Such a scenario would be analogous to how conventional T cells respond to nominal antigens. The findings that Tregs exhibit properties of effector and memory cells, as discussed above, is in agreement with this view. However, one caveat when considering Treg progression as a result of antigenic stimulation in a normal unmanipulated mouse is that the large majority of Tregs exhibit an activated phenotype, whereas most conventional T cells are resting naive T cells.

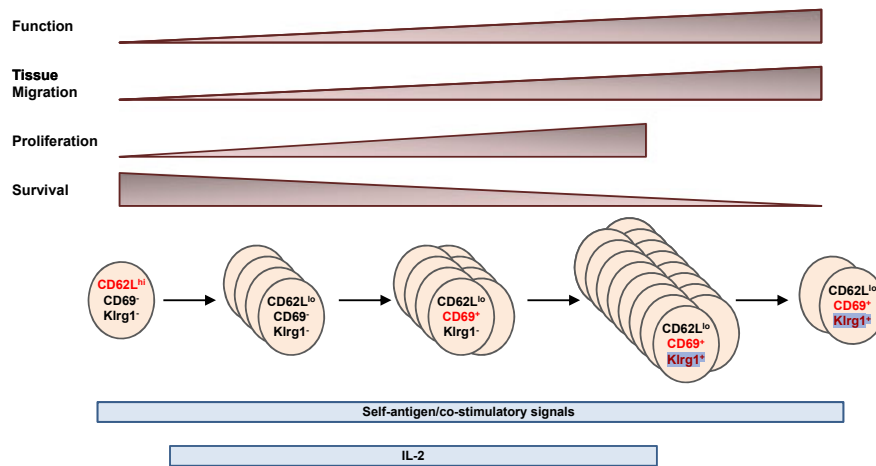


Fig. 1. Model of activation and development of Klrp1^+ Tregs.

We have initially focused on three markers which discriminate four Tregs subsets, i.e. CD62L^{hi} , CD69^- , and Klrp1^+ . When considering conventional T cells, CD62L^{hi} is represented on naive and central memory T cells, CD62L^{lo} is indicative of effector and effector memory T cells, CD69^+ represents recently activated T cells, and Klrp1^+ marks terminally differentiated Teff cells (124–126). With respect to Tregs found in the spleen and lymph nodes, Tregs can be divided into three main subsets, $\text{CD62L}^{\text{hi}} \text{CD69}^-$, $\text{CD62L}^{\text{lo}} \text{CD69}^-$, and $\text{CD62L}^{\text{lo}} \text{CD69}^+$ (127). Although there is mouse to mouse variation, generally each subset individually represents 30% of Tregs, i.e. these subsets comprise 90% of all Tregs in the spleen and lymph nodes. Approximately 10% of the Tregs in these lymphoid tissues are Klrp1^+ , and most of these are CD69^+ .

Genome-wide expression profiling was performed for these four Treg subsets as a measure of their relationship between each other. These data showed that $\text{CD62L}^{\text{hi}} \text{CD69}^-$ and $\text{CD62L}^{\text{lo}} \text{CD69}^-$ Treg subsets were more related to each other, while $\text{CD62L}^{\text{lo}} \text{CD69}^+$ and Klrp1^+ subsets were more closely related to each other (127). At one end of the spectrum was CD62L^{hi} Tregs, which exhibited properties akin to a ‘resting’ lymphoid-tissue residing subset. CD62L^{hi} Tregs expressed lower levels of proliferative markers such as Ki67, but higher levels of pro-survival molecules such as Bcl-2 (Fig. 1). At the other end of the spectrum was Klrp1^+ Tregs, which expressed higher pro-apoptotic molecules, such as caspase-1 and caspase-4, and higher levels of certain Treg suppressive molecules, e.g. granzyme B, Fgl2, IL-10, and IL-35 and chemokine receptors that favor migration into non-lymphoid tissues. We also noted that Blimp-1 expression was primarily expressed in Klrp1^+ Tregs. This raises the possibility that Blimp-1 may

either be selectively required for the development of Klrp1^+ Tregs and/or be required for the regulation of a distinctive effector program associated with these Tregs. Notably, Blimp-1 has been linked to the regulation of IL-10, a Treg suppressive molecule that is highly expressed by this Treg subset (120). The properties of these Klrp1^+ Tregs are akin to a highly activated Treg cells that are prone to apoptosis and favor immune regulation within tissue sites. Consistent with this view, CD69^+ and Klrp1^+ are dominant Treg populations in the lamina propria of the small intestine.

We developed approaches to directly follow the developmental relationship between these Tregs subsets by using adoptive transfer into lymphopenic $\text{TCR}\alpha^{-/-}$ or lymphoreplete Y3 $\text{IL-2R}\beta^{-/-}$ recipients (127). The advantage of $\text{TCR}\alpha^{-/-}$ recipients is that donor Tregs are easy to follow through their lymphopenia-driven expansion. This model, however, also requires the cotransfer of conventional Treg-depleted CD4^+ T cells as a source of growth and survival factors. An important disadvantage of $\text{TCR}\alpha^{-/-}$ recipients is that the rapid lymphopenia-induced Treg proliferation hinders tracking the progression of CFSE-labeled donor cells because CFSE is fully diluted by donor Tregs 3–4 days after transfer. However, Y3 $\text{IL-2R}\beta^{-/-}$ recipients do not have this drawback. Y3 is an $\text{IL-2R}\beta$ transgene that contains mutations in three key tyrosine residues in the cytoplasmic tail of $\text{IL-2R}\beta$ that greatly impairs IL-2 -dependent PI3K and STAT5 activation (128). When this transgene was expressed in all T-lineage cells in $\text{IL-2R}\beta^{-/-}$ mice, low IL-2 -dependent STAT5 signaling associated with the transgene supported Treg development and homeostasis but not IL-2 -dependent functions in T-effector and memory cells (128, 129). Y3 $\text{IL-2R}\beta^{-/-}$ mice are not lymphopenic, including the Treg

compartment, but in a competitive setting, wildtype donor Tregs outcompete recipient-derived Y3 $\text{IL-2R}\beta^{-/-}$ Tregs for available IL-2 . Thus, the adoptive transfer of wildtype Tregs into Y3 $\text{IL-2R}\beta^{-/-}$ recipients results in a greater accumulation of donor Tregs than occurs if wildtype mice are used as recipients. This **advantage** facilitates identification of the donor Tregs, but their accumulation is not as robust as that detected in $\text{TCR}\alpha^{-/-}$ recipients, which allows us to follow Treg subset development as cell division progresses.

Several important observations were made when CFSE-labeled wildtype 'resting' CD62L^{hi} CD69^{-} CD103^{-} Klrg1^{-} Tregs were transferred into Y3 $\text{IL-2R}\beta^{-/-}$ mice (127). **First**, by 3 days **post-transfer**, virtually all donor Tregs retained the resting phenotype and did not divide, but by 6 days **post-transfer**, the majority of the donor Tregs fully diluted CFSE, indicative of at least 7–8 cell divisions. This finding suggests that Tregs may undergo a **programming step** upon encountering self-antigen and environmental cues. Upon integration of these signals, Tregs undergo a rapid proliferative burst. **Second**, accompanying Treg proliferation, progressive loss of the resting phenotype was noted and activated Tregs accumulated, characterized by downregulation of CD62L and upregulation of CD69 , CD103 , and Klrg1 . **Third**, the appearance of CD103^{+} and especially Klrg1^{+} Tregs was observed only after at least 7–8 cell divisions. Collectively, these findings indicate that the engrafted resting Tregs respond in a lymphoreplete environment and give rise to activated Tregs in a manner analogous to conventional T cells and that expression of Klrg1 represents a **late stage** of this response (Fig. 1).

We were also interested in the question of what is the capacity of activated Tregs to adopt a resting CD62L^{hi} cell

fate. To begin to evaluate this relationship between Treg subsets, highly purified CD62L^{hi} CD69^{-} (Fr1), CD62L^{lo} CD69^{-} (Fr2), and CD62L^{lo} CD69^{+} (Fr3) Treg subsets, all depleted of Klrg1^{+} Tregs, from Foxp3/RFP reporter mice were adoptively transferred into $\text{TCR}\alpha^{-/-}$ recipients with conventional CD4^{+} T cells, which provide growth/survival signals, as discussed above. Three to four weeks **post-transfer**, donor Treg and T-conventional cells were detected in the spleen and MLN. Analysis of the spleen indicated that there was approximately a 12- to 15-fold expansion of the donor Treg and T-conventional cells (Fig. 2A). The fraction of RFP^{+} donor Tregs, therefore, still remained approximately 5–10% of the total donor pool of CD4^{+} T cells (Fig. 2B). Phenotypic analysis of the RFP^{+} Tregs revealed **heterogeneity** in expression of CD62L and CD69 for each fraction of donor Tregs, even though the input Tregs were homogenous with respect to the expression of the markers used to select the input cells (Fig. 2C, top). CD62L^{lo} CD69^{+} Tregs dominated in all recipients, including those recipients that received resting RFP^{+} CD62L^{hi} CD69^{-} (Fr1) Tregs, further supporting the notion that resting CD62L^{hi} Tregs include precursors to activated Tregs. In contrast, most 'activated' RFP^{+} Fr2 and Fr3 Tregs remained as activated cells and did not substantially move toward a resting CD62L^{hi} phenotype, suggesting that activated Tregs do not have the potential to readily develop into central memory-like cells. Donor RFP^{+} Tregs readily expressed Klrg1 in all recipients (Fig. 2C, bottom). Thus, these data also show that Klrg1^{+} Tregs readily develop from Klrg1^{-} Treg cells. A small population of resting CD62L^{hi} Tregs was detected in all recipients, which fits a scenario that a few central memory-like Tregs may also develop from activated Tregs.

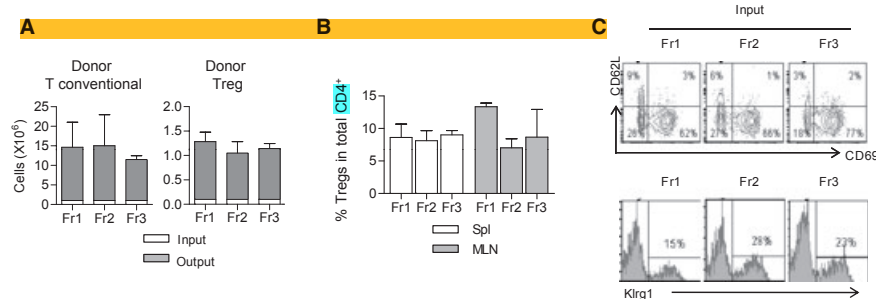


Fig. 2. Inter-relationship between Treg subsets. Adult T-cell receptor ($\text{TCR}\alpha^{-/-}$) mice were adoptively transferred with a mixture (1:10) of RFP^{+} Tregs from purified subsets (Fr1–3, Klrg1^{-}) and Treg-depleted conventional CD4^{+} T cells. The persistence and phenotype of the donor cells were assessed 3–4 weeks post-transfer. **(A)** Expansion of the donor T-conventional and Treg cells where the total number of injected (input) and recovered (output) cells from the spleen were enumerated. **(B)** Spleen and MLN were examined for donor Tregs as a fraction of the total CD4^{+} T-cell compartment. **(C)** The phenotype of donor Tregs from recipients that received the indicated fractions of Klrg1 -depleted Fr1, Fr2, or Fr3 RFP^{+} donor Treg cells. Shown at the top is the expression of CD62L and CD69 and at the bottom is the expression of Klrg1 by donor Tregs.

Expression of **Klrg1** marks short-lived terminally differentiated Tregs

Klrg1 has been primarily studied in NK and CD8⁺ T cells. **Klrg1** is detected on these cells only after they have undergone extensive proliferation and marks cells with replicative senescence (125, 126, 130–132). **Klrg1** is associated with mature NK cells with poor proliferative potential that arise from **Klrg1**[−] NK cells (133, 134). During a CD8⁺ T-cell immune response, **Klrg1** expression is associated with short-lived highly activated Tregs (125, 126, 132). Analogous to **Klrg1**⁺ CD8⁺ T cells, **Klrg1**⁺ Tregs have a gene expression profile that is consistent with cells that exhibit highly suppressive, proliferative, and apoptotic activity. Moreover, we performed direct studies to test whether **Klrg1**⁺ Tregs are indeed short-lived terminally differentiated Tregs (127). First, **Klrg1**⁺ Tregs turnover faster than **Klrg1**[−] Tregs in vivo as assessed by BrdU-labeling experiments and exhibited increased suppressive activity in vitro. Second and more importantly, **Klrg1**⁺ Tregs did not expand and poorly survived upon adoptive transfer into **TCRα**^{−/−} or Y3 **IL-2Rβ**^{−/−} recipients, indicating that these Tregs are short-lived, even though they show high proliferative levels at the steady state. For the few **Klrg1**⁺ Tregs detected in these recipients, they remained **Klrg1**⁺ and stably expressed **Foxp3**. These data indicate that once **Klrg1** is expressed on Tregs, most of these cells quickly become exhausted and are in a terminally differentiated state. Collectively, the late developmental progression, the gene expression profile, the preference for non-lymphoid tissue residence, and these direct studies indicate that **Klrg1** marks highly functional, short-lived terminally differentiated Tregs that act predominantly to maintain tolerance and homeostasis in non-lymphoid tissues.

Signaling through the **IL-2R** represents one important requirement for the development of **Klrg1**⁺ Tregs. This was first suggested from genome-wide profiling of splenic Tregs with normal and impaired **IL-2R** signaling, where the level of **Klrg1** was one of the most differentially and highly expressed mRNAs by wildtype Tregs (128). Analysis of Tregs with impaired **IL-2R** signaling showed that **Klrg1**⁺ Tregs were highly diminished. This finding was most striking for the lamina propria of the small intestine, where **Klrg1**⁺ Tregs comprise approximately 50% of the Tregs in wildtype mice but only about 5% in **IL-2R** signaling impaired mice (127). Interestingly, direct assessment of the effect of

agonist **IL-2**/anti-**IL-2** complex on **Klrg1**⁺ and **Klrg1**[−] Tregs showed that only the latter cells respond to **IL-2**. Thus, **IL-2** acts on **Klrg1**[−] cells to drive the development of **Klrg1**⁺ Tregs, but the **Klrg1**⁺ Tregs remain senescent and non-proliferative to **IL-2**. This finding also has important implications concerning the mechanisms in play with respect to **IL-2**-dependent Treg homeostasis (Fig. 1). Recent studies indicate that expression of prosurvival Mcl-2 by **IL-2** contributes to Treg homeostasis (135). However, **IL-2**-dependent development of **Klrg1**⁺ Tregs is also likely an important control point for Treg homeostasis. **Klrg1**[−] Tregs respond to **IL-2**, which contributes to their proliferation. After perhaps >8 cell divisions, Tregs progress and develop into **Klrg1**⁺ cells. The **Klrg1**⁺ Tregs quickly cease to proliferate, are non-responsive to **IL-2**, and will die in a short-period of time. Thus, even though Tregs are highly proliferative cells in vivo at the steady state, the death of **Klrg1**⁺ Tregs provides a mechanism to link **IL-2R** signaling to limit and balance Treg numbers, maintaining homeostasis.

Concluding remarks

Current studies make plain that **Foxp3**⁺ Tregs in the periphery are mostly activated T cells that exhibit substantial heterogeneity in a manner very analogous to developing and polarized Th subsets. Treg functional and migratory diversity provides a mechanism to optimally match suppressive Tregs to particular T-effector responses. Tregs also exhibit diversity that in part reflects the response to self and nominal antigenic stimulation and other signals to select Tregs with appropriate specificities and expand them to sufficient numbers to regulate T-effector responses. This process requires mechanisms to limit the Treg response, such as development of **Klrg1**⁺ Tregs, and produce persistent memory Tregs. Both are important for immune homeostasis, as the former promotes Treg homeostasis and the latter promotes homeostasis of the T-conventional compartment. With this understanding, there remain open a number of key questions. Several that we consider are as follows: what signals promote Treg effector versus memory responses, to what degree are resting and activated Treg subsets related to each other or represent specialized subsets of cells, and to what extent does some of the heterogeneity of Tregs defined using sophisticated mouse models apply to the human immune system?

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